

# Linux Directory Layout Cheat Sheet

What lives where on a Linux system, so you know where to find configs, logs, data and programs.

## The tree

|                    |                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------|
| /                  | ← the root of everything; every disk is mounted somewhere below it                          |
| -- bin/ sbin/ lib/ | ← commands, admin commands, libraries. Today just symlinks into /usr/                       |
| -- usr/            | ← installed software; packages own it, treat as read-only                                   |
| -- bin/ sbin/      | ← almost all programs (python3, git, nginx)                                                 |
| -- lib/            | ← libraries; lib/systemd/system/ = packaged units; lib/modules/ = drivers                   |
| -- share/          | ← docs, man pages, icons, default data                                                      |
| `-- local/         | ← things <b>you</b> install by hand (local/bin); package manager leaves it alone            |
| -- etc/            | ← system-wide config: plain text files (see below)                                          |
| -- var/            | ← data that changes while running                                                           |
| -- log/            | ← log files (syslog or messages, nginx/, journal/)                                          |
| -- lib/            | ← app state: databases (mysql/, postgresql/), docker/, libvirt/, package DBs                |
| -- cache/          | ← downloads and caches that can be rebuilt (apt, dnf)                                       |
| `-- spool/ tmp/    | ← queues (mail, users' crontabs) · temp files that survive a reboot                         |
| -- home/ root/     | ← users' home directories (/home/mark) · root's home is /root                               |
| -- opt/ srv/       | ← self-contained third-party apps (/opt/google/chrome) · data this server serves (websites) |
| -- tmp/            | ← scratch space for everyone; <b>cleared at boot</b> on many systems                        |
| -- boot/           | ← kernel (vmlinuz-*), initramfs, GRUB config; keep free space here                          |
| -- dev/            | ← device files: disks, terminals, null (see below)                                          |
| -- proc/ sys/      | ← live kernel and process info · hardware and driver tree. Both virtual: nothing on disk    |
| -- run/            | ← PID files and sockets for running services; in RAM, wiped at boot                         |
| `-- mnt/ media/    | ← your temporary manual mounts · USB sticks auto-mounted (/media/mark/USB)                  |

## Notable files in /etc

| File                   | Holds                                      | File                       | Holds                               |
|------------------------|--------------------------------------------|----------------------------|-------------------------------------|
| <b>fstab</b>           | Disks to mount at boot                     | <b>systemd/system/</b>     | Your unit files and overrides       |
| <b>hosts</b>           | Local name-to-IP lookups                   | <b>netplan/*.yaml</b>      | Network config (Ubuntu)             |
| <b>passwd, group</b>   | User accounts ( <b>shadow</b> : passwords) | <b>crontab, cron.d/</b>    | System-wide cron jobs               |
| <b>ssh/sshd_config</b> | SSH server settings                        | <b>sudoers.d/</b>          | sudo rules; edit with <b>visudo</b> |
| <b>os-release</b>      | Distro name and version                    | <b>apt/ / yum.repos.d/</b> | Package repos (Debian / RHEL)       |

## Inside /dev and /proc

| Path                      | Is                                  | Path                 | Is                                    |
|---------------------------|-------------------------------------|----------------------|---------------------------------------|
| <b>/dev/sda, sda1</b>     | SATA/USB disk 1 · its partition 1   | <b>/proc/cpuinfo</b> | CPU model, cores, flags               |
| <b>/dev/nvme0n1p1</b>     | NVMe disk 0, partition 1            | <b>/proc/meminfo</b> | Detailed memory numbers               |
| <b>/dev/null</b>          | Black hole: discards what you write | <b>/proc/1234/</b>   | One dir per PID: cmdline, status, fd/ |
| <b>/dev/zero, urandom</b> | Endless zero bytes / random bytes   | <b>/proc/sys/</b>    | Kernel tunables (use <b>sysctl</b> )  |

## Where do I look for...

| Looking for                  | Look in                                                                                                    |
|------------------------------|------------------------------------------------------------------------------------------------------------|
| An app's config              | <b>/etc/&lt;app&gt;/</b> (e.g. <b>/etc/nginx/</b> ); your overrides often go in a <b>conf.d/</b> subfolder |
| Logs                         | <b>/var/log/</b> and <b>journalctl -u &lt;service&gt;</b>                                                  |
| App data (databases, Docker) | <b>/var/lib/&lt;app&gt;/</b> (e.g. <b>/var/lib/mysql/</b> , <b>/var/lib/docker</b> )                       |
| Where a program is installed | <b>command -v nginx</b> ; then <b>dpkg -S rpm -qf</b> on that path for its package                         |
| Service unit files           | <b>/etc/systemd/system/</b> (yours, wins) · <b>/usr/lib/systemd/system/</b> (packages)                     |
| Cron jobs                    | <b>crontab -l</b> · <b>/etc/crontab</b> · <b>/etc/cron.d/</b> · <b>/var/spool/cron/</b>                    |
| What's filling the disk      | <b>sudo du -xh -d1 /   sort -h</b> (-x stays on one filesystem)                                            |
| A user's settings (dotfiles) | <b>~/.bashrc</b> , <b>~/.profile</b> , <b>~/.ssh/</b> , <b>~/.config/</b> ( <b>ls -a</b> shows them)       |

**Remember:** config in **/etc**, data in **/var**, your stuff in **/home** or **/srv**. Don't hand-edit **/proc** or **/sys** unless you know why; changes there take effect instantly and vanish at reboot.